

Functional Requirements Specification (FRS)

Project Name: Web Crawl, Accessibility & Compliance Monitoring System

Version: v0.1

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Date: 10-02-2026

Status: Draft

REVISION HISTORY

Version	Date	Author	What Changed
v0.1	10-02-2026	Sarthak Mishra	Initial Draft
v1.0			

1. PROJECT OVERVIEW

1.1 What Are We Building?

The Web Crawl, Accessibility & Compliance Monitoring System is a web-based platform that continuously crawls registered websites to discover pages, analyze accessibility and compliance issues, and generate actionable reports. It is designed for engineering, compliance, and audit teams to gain real-time visibility into website health without relying on manual checks or spreadsheets. The system replaces periodic, manual audits with automated, scalable, and audit-ready monitoring. It supports historical tracking, trend analysis, and regulator-ready exports.

1.2 Who Are the Users?

Role	What They Do
Viewers	Views dashboards, reports, and issue summaries
Developer	Reviews issues, updates statuses, adds notes
Admin	Manages teams, sites, crawl settings, and access

1.3 What Are We NOT Building? (Out of Scope)

- No mobile application (web-only)
- No real-time blocking or auto-fixing of issues
- No payment or billing module in v1.0

2. MODULES & FEATURES

M = Must Have | **S** = Should Have | **C** = Could Have | **W** = Won't Have (this version)

Module	Feature	Priority
Team management	Create And Manage Teams	M
Site Management	Register websites	M
Site Management	Activate/Deactivate Monitoring	M
Crawl Management	Trigger Crawl Jobs	M
Crawl Management	Automated Crawl Scheduling	S
Crawl Engine	Page discovery & crawling	M
Analysis Engine	Accessibility analysis (axe-core)	M
Analysis Engine	Compliance rule checks	M
Issue Management	View detected issues	M
Issue Management	Update issue status	M
Issue Management	Add review Notes	S
Reporting	Daily Site Metrics	M
Reporting	Compliance & accessibility reports	S
Observability	Crawl Job Status Tracking	M

3. FUNCTIONAL REQUIREMENTS

FR-001 — Team Creation & Management

Field	Details
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ID	FR-001
Feature	Team Management
Module	Team Management
Priority	Must have
Who uses It	Admin
What Does It Do	The system shall allow admins to create and manage teams that own websites.
Before It Runs	Admin is authenticated
After it Runs	Team is created and available for site assignment
API Endpoint	POST /api/teams

It works correctly when:

- Admin creates a team with a unique name → team is created
- Duplicate team name → validation error
- Missing required fields → error message returned

FR-002 — Website Registration

Field	Details
ID	FR-002
Feature	Website registration
Module	Site Management
Priority	Must have
Who uses It	Admin
What Does It Do	The system shall allow admins to register a website under a team for monitoring
Before It Runs	Team exists
After it Runs	Website is stored and ready for crawl
API Endpoint	POST /api/sites

It works correctly when:

- site details entered → site is registered
- Invalid base URL → validation error

- Team does not exist → error returned

FR-003 — Crawl Job Triggering

Field	Details
ID	FR-003
Feature	Crawl Job Trigger
Module	Crawl Management
Priority	Must have
Who uses It	Admin
What Does It Do	The system shall allow users to trigger a crawl job for a site.
Before It Runs	Site Is Active
After it Runs	Crawl Job is created with pending status API
API Endpoint	POST /api/crawl-jobs

It works correctly when:

- Crawl triggered → job created
- Site inactive → crawl blocked
- Duplicate triggers → separate jobs created

FR-004 — Page Discovery & Crawling

Field	Details
ID	FR-004
Feature	Page Crawling
Module	Crawl Engine
Priority	Must have
Who uses It	System
What Does It Do	The system shall discover and crawl pages using background workers.
Before It Runs	Crawl Job Exists
After it Runs	Pages and page versions are stored
API Endpoint	Internal Worker Process

It works correctly when:

- Pages are discovered and queued
- Failed pages are retried
- Crawl continues despite partial failures

FR-005 — Accessibility Analysis

Field	Details
ID	FR-005
Feature	Accessibility Checks
Module	Analysis Engine
Priority	Must have
Who uses It	System
What Does It Do	The system shall analyze crawled pages for accessibility issues using axe-core.
Before It Runs	Page Version Exists
After it Runs	Issue Instances are recorded
API Endpoint	Internal Worker Process

It works correctly when:

- Accessibility issues are detected and stored
- Multiple issues can exist per page version
- Analysis failures do not block crawling

FR-006 — Issue Status Management

Field	Details
ID	FR-006
Feature	Issue Lifecycle
Module	Issue Management
Priority	Must have
Who uses It	Developer

What Does It Do	The system shall allow users to update the status of an issue.
Before It Runs	Issue Instance Exists
After it Runs	Status History recorded
API Endpoint	POST /api/issues/{id}/status

It works correctly when:

- Status change is saved
- History is preserved
- Unauthorized users are blocked

FR-007 — Reporting & Metrics

Field	Details
ID	FR-007
Feature	Reporting
Module	Reporting
Priority	Should Have It
Who uses It	Viewer
What Does It Do	The system shall generate daily metrics and compliance reports per site.
Before It Runs	Crawl Data Exists
After it Runs	Reports Are Available for Viewing
API Endpoint	GET /api/reports

It works correctly when:

- Daily metrics are generated
- Historical trends are visible
- Report exports match stored data

4. NON-FUNCTIONAL REQUIREMENTS

ID	Category	Requirement
NFR-01	Performance	Dashboard Loads Under 2 sec
NFR-02	Performance	Web crawling and analysis must not block user interactions
NFR-04	Security	Role Based Access Control
NFR-05	Observability	Job Status And failures are visible
NFR-06	Maintainability	Modular microservices architecture
NFR-07	Asynchronous processing	All crawling and analysis must run in background processes
NFR-08	Scalability	All crawling and analysis must run in background processes

5. DATA REQUIREMENTS

Main entities:

- Users

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null
email	string	yes	Not null
passwordHash	string	yes	Not null
displayName	string	yes	Not null
role	string	yes	Viewer or Developer or Admin
createdAt	timestamp	yes	Auto-set

- Teams

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null

name	string	yes	Unique, not null
description	string	no	
createdAt	timestamp	yes	Auto-set

- Sites

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null
team_id	uuid	yes	Not null
name	string	yes	Not null
baseURL	string	yes	Not null
isActive	string	yes	Yes or No
createdAt	timestamp	yes	Auto-set

- Crawl Jobs

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null
site_id	uuid	yes	Not null
status	string	yes	Not null
trigger_type	string	yes	Not null
requested_by	uuid	yes	Not null
started_at	timestamp	yes	Not null
completed_at	timestamp	yes	Not null
created_at	timestamp	yes	Auto-set

- Pages

Field	Type	Required	Notes
site_id	uuid	yes	Not null
url	string	yes	Not null
first_discovered_at	timestamp	yes	Not null
last_seen_at	timestamp	yes	Not null
id	uuid	yes	Primary key, not null
status	text	yes	Not null

- Page Versions

Field	Type	Required	Notes
page_id	uuid	yes	Not null
id	uuid	yes	Primary key, not null
crawl_job_id	uuid	yes	Not null
http_status	int	yes	Not null
content_hash	text	yes	Not null
title	text	yes	Not null
content_size	int	no	
created_at	timestamp	yes	Auto-set

- Issue Definitions

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null
rule_code	text	yes	unique
name	text	yes	Not null
description	text	no	
severity	text	yes	Not null
wcag_reference	text	yes	Not null

- Issue Instances

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null
page_version_id	uuid	yes	Not null
issue_definition_id	uuid	yes	Not null
element_selector	text	no	
message	text	no	
impact	text	yes	Not null
first_detected_at	timestamp	yes	Not null

- Issue Status History

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null

issue_instance_id	uuid	yes	Not null
previous_status	text	no	
new_status	text	yes	Not null
changed_by	uuid	yes	Not null
changed_at	timestamp	yes	Not null
note	text	no	

- Site Daily Metrics

Field	Type	Required	Notes
id	uuid	yes	Primary key, not null
site_id	uuid	yes	Not null
metric_date	date	yes	Not null
pages_scanned	int	yes	Default 0
total_issues	int	yes	Default 0
critical_issues	int	yes	Default 0
serious_issues	int	yes	Default 0
moderate_issues	int	yes	Default 0
minor_issues	int	yes	Default 0
accessibility_score	decimal	yes	Default 0.0

6. EXTERNAL INTEGRATIONS

Service	Purpose
Puppeteer	Rendering And Crawling
Axe-core	Accessibility analysis
RabbitMQ	Asynchronous task processing
Redis	Caching
PostgreSQL	Primary Data Storage

7. ASSUMPTIONS

- Websites allow crawling
- Updation and assignment of roles is done by DBA

- Users will access the system via modern browsers (Chrome, Firefox, Safari, Edge)
- The system will be used in English only
- Target websites allow automated crawling and are not blocked by bot protection mechanisms

8. OPEN QUESTIONS

Question	Asked By	Answer	Status
Should reports be exportable as PDF in v1.0?	Sarthak Mishra		Open
Should crawl schedules be configurable per site?	Sarthak Mishra		Open

SIGN-OFF

Role	Name	Date	Signature
Product Owner			
Developer/ Tech Lead			
Product Manager			